Tools

Overview

Analyzing questions related to tool use.

Research questions: - How are OSPOs contributing to their org's open source tooling and infrastructure?
- How are OSPOs engaging with AI, as advisors?
- How are OSPOs engaging with AI, as users?
- What tools do OSPOs rely on?
- Is available tooling a limitation for OSPOs?

Survey questions: - Supporting tools and infrastructure (tools_infra)
- Which AI-related topics or services has your OSPO worked on? (ai_topics)
- How does your OSPO use AI tools in its own work? (ai_usage)
- Which of the following does your OSPO rely on in its day-to-day work? (tools_reliance)

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tibble)
```

```
Warning: package 'tibble' was built under R version 4.4.3
```

```
library(docstring)
```

```
Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
codebook <- read_excel(file.path(cfg$data_path, "codebook.xlsx"))
tools_infra <- read_tsv(file.path(cfg$data_path, "tools_infra.tsv"), show_col_types = FALSE,
ai_topics <- read_tsv(file.path(cfg$data_path, "ai_topics.tsv"), show_col_types = FALSE, ski
ai_usage <- read_tsv(file.path(cfg$data_path, "ai_usage.tsv"), show_col_types = FALSE, skip_
tools_reliance <- read_tsv(file.path(cfg$data_path, "tools_reliance.tsv"), show_col_types =
blockers <- read_tsv(file.path(cfg$data_path, "blockers.tsv"), show_col_types = FALSE, skip_
matrix_other <- read_tsv(file.path(cfg$data_path, "matrix_other.tsv"), show_col_types = FALS
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, s
```

How are OSPOs contributing to their org's open source tooling and infrastructure?

First we need to, once again, filter responses to one response per OSPO.

```
tools_infra$organization <- org_responses$organization

dupes <- tools_infra |>
  group_by(organization) |>
  filter(n() > 1) |>
  arrange(organization, .by_group = TRUE)

# Keep commented for privacy!!
# dupes
```

UGH this is super annoying. There is substantial disagreement. For example, when asked whether they provide or manage code hosting, every single organization that gave >1 response had disagreement among the responses. So I have some options for how to proceed:
1) Choose one respondent at random.
2) Choose one respondent at random, and then run several versions of the analysis with different randomly selected "respresentatives", and see how much the final results change among the different combinations.
3) Select "yes" as the institution's answer if ANYBODY from that institution said "yes".

Obviously it's very imperfect, but I'm going to go with #3. It seems more robust/justifiable than #1, and I don't really have time to do #2. I've added a function to utils.R to consolidate the df for these SATA questions according to the heuristic in #3.

```
tools_infra_consolidated <- consolidate_org_responses_yn(tools_infra)
```

I've looked at the results manually and the function seems to work as it should.

Okay, so now we can actually proceed to actually draw some conclusions from the data.

```
count_yes_responses(tools_infra_consolidated)
```

```
# A tibble: 8 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    15 Maintaining a catalog or registry of the institution's open so~ catalog
2    14 Providing or managing code hosting (for example an institution~ code_ho~
3     9 <NA>                                                          metrics
4     9 Providing metrics or dashboards on open source activity       metrics
5     8 Automating checks for licensing, security, or good practice   checks_~
6     8 Building Open Source software to support research             build_r~
7     7 Providing development infrastructure such as CI/CD or testing ~ dev_inf~
8     4 Building other Open Source software (e.g. EdTech)             build_o~
```

So it looks like the most common ways that OSPOs contribute to their organization's tooling/infrastructure for open source are through software catalogs and code hosting. Building EdTech software is the least common.

How are OSPOs engaging with AI, as advisors?

```
ai_topics$organization <- org_responses$organization
ai_topics_consolidated <- consolidate_org_responses_yn(ai_topics)
count_yes_responses(ai_topics_consolidated)
```

```
# A tibble: 7 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    15 Contributing to institutional or sector AI policy discussions ai_poli~
2    12 Running training or awareness sessions on AI and open source  ai_trai~
3    12 Supporting the open release of models, weights, or training da~ ai_mode~
4    12 Helping researchers evaluate or adopt open source AI tools    ai_eval~
5    11 Advising on licensing or compliance for AI models, weights, or~ ai_mode~
6    10 Advising on licensing of AI-generated code                    ai_lice~
7     5 Developing institutional guidance on the use of AI coding assi~ ai_codi~
```

So it looks like the most common answer is "Contributing to institutional or sector AI policy discussions". Other popular answers include "Running training or awareness sessions on AI and open source", "Supporting the open release of models, weights, or training datasets", and "Helping researchers evaluate or adopt open source AI tools". "Developing institutional guidance on the use of AI coding assistants" was the least common response.

How are OSPOs engaging with AI, as users?

For this one, let's not consolidate by org since AI usage is largely an individual choice.

```
# ai_usage$organization <- org_responses$organization
# ai_usage_consolidated <- consolidate_org_responses_yn(ai_usage)
# count_yes_responses(ai_usage_consolidated)
```

```
# Count rows where ai_use_none OR ai_use_avoid is "Yes"
count <- sum(ai_usage$ai_use_none == "Yes" | ai_usage$ai_use_avoid == "Yes", na.rm = TRUE)
(nrow(ai_usage) - count) / nrow(ai_usage)
```

```
[1] 0.9090909
```

^ This is the percent of respondents who do not use AI at all in their daily work

What tools do OSPOs rely on?

```
tools_reliance$organization <- org_responses$organization
tools_reliance_consolidated <- consolidate_org_responses_yn(tools_reliance)
count_yes_responses(tools_reliance_consolidated)
```

```
# A tibble: 9 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    22 A website or intranet presence for the OSPO                   tools_w~
2    20 Code hosting platforms (for example GitHub, GitLab)           tools_c~
3    18 Chat platforms (for example Slack, Teams, Zulip)              tools_c~
4    17 Mailing lists                                                 tools_e~
5    12 Document templates (for example charters, license guidance, co~ tools_t~
6    11 A project catalog or registry                                 tools_c~
```

```
7    10 A ticketing or request system for handling inquiries            tools_t~
8     8 Existing institutional events and gatherings (for example town~ tools_e~
9     5 Metrics dashboards or reporting tools                           tools_m~
```

Is tooling a blocker for OSPOs?

```
score_map <- c(
  "Not a blocker for us"  = 0L,
  "Minor, we work around it"   = 1L,
  "Moderate, it slows us down" = 2L,
  "Major, it limits our impact"        = 3L
)

# Convert all text columns to numeric values
blockers_numeric <- blockers |>
  mutate(across(everything(), ~ score_map[.x]))

blockers_numeric$organization <- org_responses$organization
blockers_numeric_consolidated <- consolidate_org_responses_num(blockers_numeric)
```

```
tally_matrix_vals_by_org(blockers_numeric_consolidated)
```

```
# A tibble: 11 x 3
   count tip                                                          nickname
   <int> <chr>                                                        <chr>
 1    25 Limited staff time or funding for the function itself        resourc~
 2    24 Limited awareness of open source practices among researchers ~ awarene~
 3    23 Short-term funding cycles versus long-term sustainability nee~ funding~
 4    22 Difficulty demonstrating impact or value                     trackin~
 5    21 Lack of visible senior leadership sponsorship                sponsor~
 6    21 Tooling or infrastructure constraints                        tools
 7    20 Developing institutional policy and strategy for open source policy
 8    20 Policies designed for proprietary or commercialization contex~ policy
 9    18 Legal, compliance, or security processes that slow open sourc~ complia~
10    18 Incentive structures that do not reward open source work (for~ incenti~
11    17 Cultural resistance to sharing or openness                   culture
```

Write-in responses.

Finally, let's look at the comments people wrote in regarding tools. If I judge that we are getting some interesting content from a particular question, I may write it to its own file for analysis in QualCoder.

```
# Keep commented for privacy!!

# sata_other$tools_infra_other
# sata_other$tools_other
# sata_other$ai_topics_other
# sata_other$ai_use_other
```

Meh, I don't see anything super compelling here.